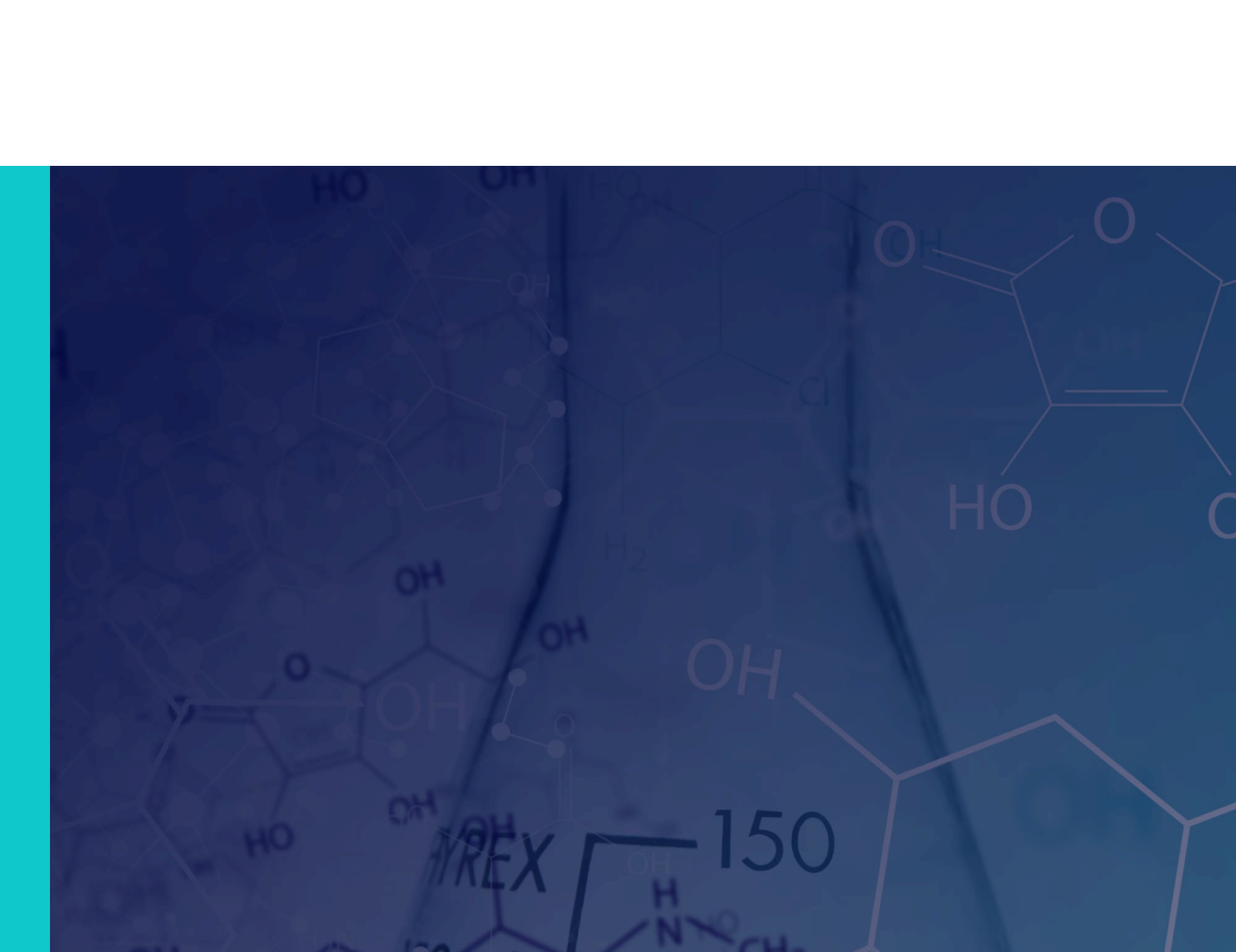




McGill CHEM 120

Summer 2025, Chapter 7 Notes



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7. Stoichiometry [review]

7.1 Introduction to Chemical Equations and Balancing

7.1.1

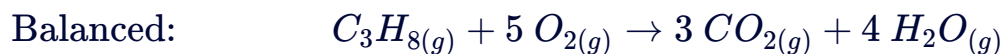
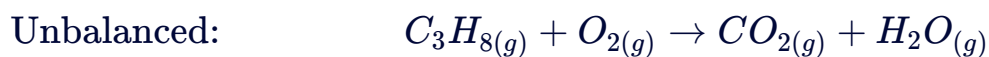
Balancing Reactions

Reactions must be balanced before the stoichiometry can be analyzed.

What makes a reaction balanced?

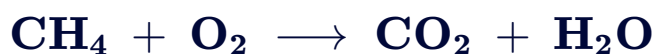
The number of _____ of each type on the _____ must equal the number of atoms of each type on the _____ of the reaction.

Example:



Balancing Chemical Reactions

Unbalanced Reaction



The **subscripts** tell us the ratio of different atoms in one molecule.

Example: One CH_4 molecule has 4 hydrogen atoms and 1 carbon atom

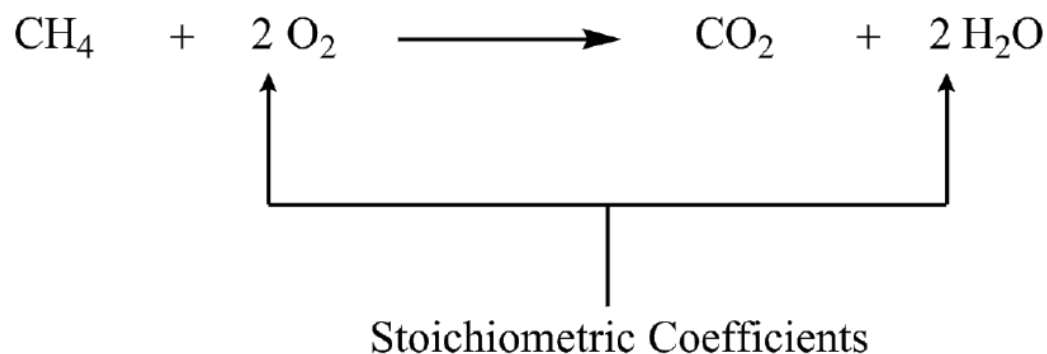
WATCH OUT!

Sometimes you could be given unbalanced equations and you are expected to balance them!
We **need to use the balanced equation for calculations** in order to get the correct answer!

Steps to Balance Chemical Equations

1. Balance **elements that only appear once** on the reactant side and once on the product side (typically elements not including C, N, H, O)
2. **Group polyatomic ions** and balance them as a group (not as separate elements, e.g. balance NO_3 as 1 NO_3 group, not 1 N and 3 O)
3. **Balance all other elements**, starting with the least common elements to the most common elements
4. Make sure all **coefficients are whole numbers** AND are represented in the **simplest integer ratio possible**
5. **Check** to make sure all elements are balanced on each side of the chemical equation

Balanced Reaction



- The **stoichiometric coefficients** of the balanced equation tell us that we need _____ oxygen molecules for each methane (CH_4) molecule and we produce _____ waters for every methane consumed.
- It also tells us that for every 1 mole of methane used we produce _____ mole of CO_2 .

WIZE CONCEPT

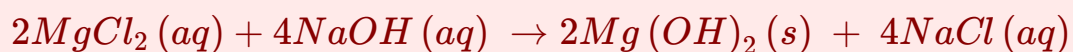
Multiply coefficients by subscripts to see exactly how much of an atom we have on each side.

! WATCH OUT!

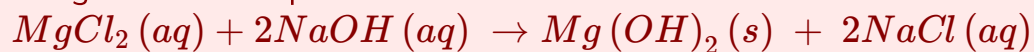
We want the **lowest possible coefficients**.

Example:

If I balanced an equation and got:



The **greatest common factor** that can go into each coefficient is 2 so we need to divide each coefficient by 2 to get the lowest possible numbers!



Example: Balancing a Chemical Reaction

Balance the following equation:



WIZE CONCEPT

The physical state of each chemical compound is often included in chemical equations:

(s) = solid

(l) = liquid

(g) = gas

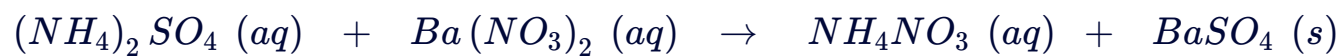
(aq) = aqueous

Polyatomic Ions

Note: There is no need to memorize the following table. It is just included here so you have an idea about what a polyatomic ion is!

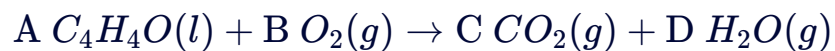
Polyatomic Ion Cheatsheet			
Ion	Name	Ion	Name
NH_4^+	Ammonium	MnO_4^-	Permanganate
NO_2^-	Nitrite	CrO_4^{2-}	Chromate
NO_3^-	Nitrate	$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
SO_3^{2-}	Sulfite	O_2^{2-}	Peroxide
SO_4^{2-}	Sulfate	O_2^-	Superoxide
HSO_4^-	Hydrogen sulfate [*]	$\text{C}_2\text{O}_4^{2-}$	Oxalate
HSO_3^-	Hydrogen sulfite	HC_2O_4^-	Hydrogen Oxalate
OH^-	Hydroxide	HCO_2^-	Formate
CN^-	Cyanide	$\text{S}_2\text{O}_3^{2-}$	Thiosulfate
PO_4^{3-}	Phosphate	HS_2O_3^-	Hydrogen Thiosulfate
HPO_4^{2-}	Hydrogen Phosphate	BrO_4^-	Perbromate
H_2PO_4^-	Dihydrogen Phosphate	BrO_3^-	Bromate
CO_3^{2-}	Carbonate	BrO_2^-	Bromite
HCO_3^-	Hydrogen Carbonate ^{**}	BrO^-	Hypobromite
ClO^-	Hypochlorite	IO_4^-	Periodate
ClO_2^-	Chlorite	IO_3^-	Iodate
ClO_3^-	Chlorate	IO_2^-	Iodite
ClO_4^-	Perchlorate	IO^-	Hypoiodate
$\text{C}_2\text{H}_3\text{O}_2^-$	Acetate		

Example: Balance a Chemical Equation with Polyatomic Ions



Practice: Balancing the Combustion of Furan

Choose the answer that balances the reaction shown below:



A = 2; B = 9 ;C = 8; D = 4

☐

A = 1; B = 3 ;C = 5; D = 2

☐

A = 1; B = 1 ;C = 1; D = 1

☐

A = 4; B = 25 ;C = 20; D = 10

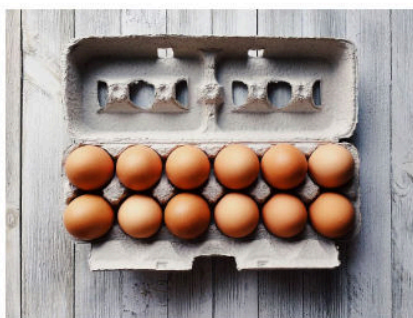
☐

7.2 Atoms, Molecules, and the Mole!

7.2.1

Moles

When someone says they want a dozen donuts, roses, or eggs we know they mean they want 12. A **mole** in chemistry also tells us the amount of something.



- Just like how a dozen of something = 12, 1 mole of something = 6.02×10^{23} molecules
- 6.02×10^{23} is referred to as **Avagadro's number** (N_A)

i WIZE TIP

You should memorize Avagadro's number since it is not always provided on exams! We'll soon see that you need to know it to solve problems :)

$$N_A = 6.02 \times 10^{23} \text{ molecules/mole}$$

We use the unit "**moles**" to help make very small amounts more measurable. A mole is just like any other unit!

There are 2 Equations Related to Moles:

$$n = \frac{m}{M}$$

$$n = \frac{N}{N_A}$$

n =# of moles

m =mass (g)

M =molar mass (g/mol)

N =# of molecules or atoms

N_A =Avagadro's number= 6.02×10^{23} molecules

Example #1:

If we are told that a sample of $\text{CO}_2(\text{s})$ weighs 11g, how many moles are present?

Example #2

We have 2 moles of CO_2 present in our sample, how many molecules are there?

Example #3

How many oxygen atoms there in 1 mole of CO_2 ?

1) Find Number of Molecules In the Sample

2) Find the Number of O atoms in the Sample

In each molecule there are _____ O atoms.

Therefore to find the number of O atoms:

of molecules in sample x _____ O atoms/molecule = # of O atoms

of O atoms = _____

Example: Calculate the Number of Molecules of Ethyl Mercaptan

The volatile liquid ethyl mercaptan, $\text{C}_2\text{H}_6\text{S}$, is one of the most odoriferous substances known. It is added to natural gas to make gas leaks detectable. ($d = 0.84 \text{ g/mL}$; $\text{MW} = 62.1 \text{ g/mol}$)

1. How many $\text{C}_2\text{H}_6\text{S}$ molecules are contained in a $3.0 \mu\text{L}$ sample?

2. In the same 3.0 μL sample, how many C and H atoms are there?

Practice: Converting Mass to Number of Atoms

Calculate the number of nitrogen atoms in 2.25 g of bismuth(III) nitrate.

a) 1.03×10^{22} atoms

☐

b) 1.03×10^{21} atoms

☐

c) 3.43×10^{21} atoms

☐

d) 3.43×10^{22} atoms

☐

Practice: Finding the Number of Moles of Iron

Calculate the number of moles of iron atoms in 14.1 g of iron oxide, Fe_2O_3

A) 0.177 mol

☐

B) 0.0821 mol

☐

C) 0.0906 mol

☐

D) 0.0451 mol

☐

7.3 Stoichiometry

7.3.1

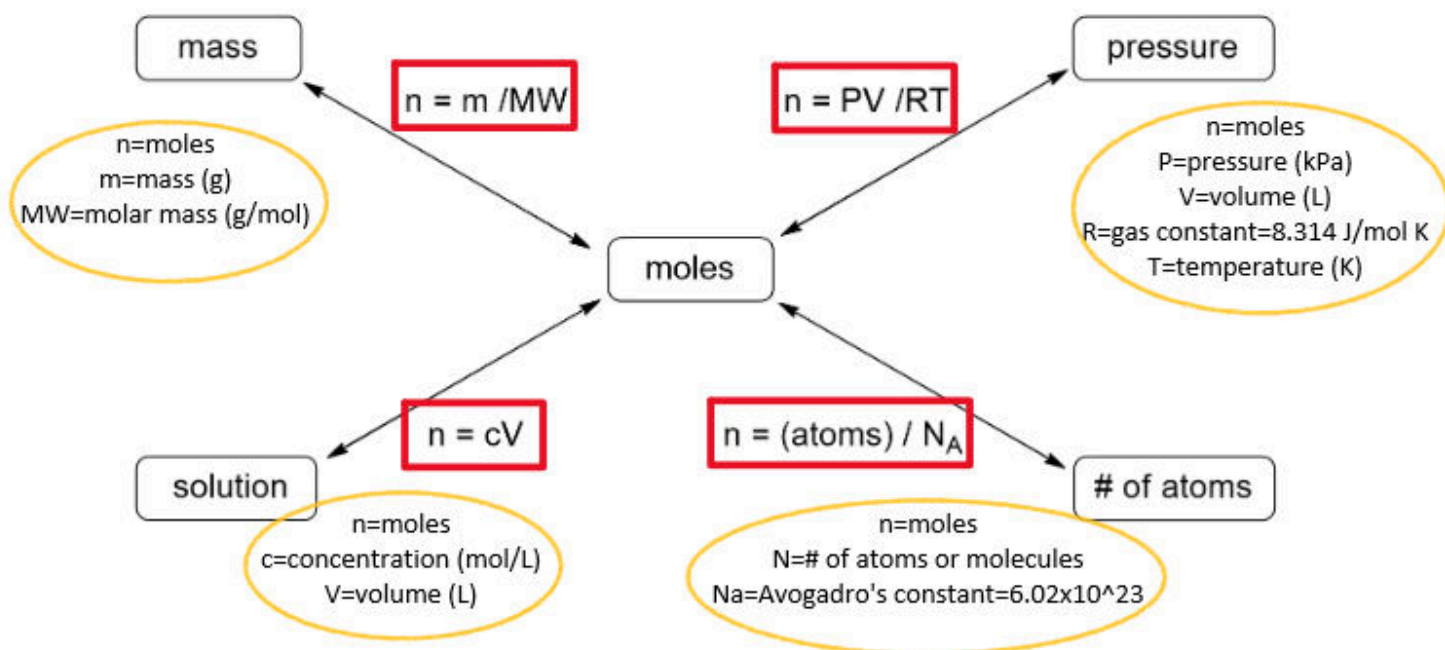
Intro to Stoichiometry

Stoichiometry allows us to predict the quantities of products or reagents across a chemical reaction.

Typically this is done in a few steps:

1. Calculating the number of moles of the reagents
2. Calculating the number of moles of the products using the stoichiometric ratio
3. Calculating the mass or pressure of the products.

Moles are the central unit!



In addition we can convert between mass and volume of a pure substance by looking at it's **density**,

$$\delta = m/V$$

Stoichiometry of a Reaction

We use the **coefficients** of the **balanced reaction** along with the our equations that convert mass, volume, and concentration into moles to predict the quantities of reactants and products in a chemical reaction.

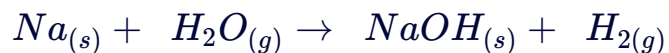
To answer any stoichiometry problem, focus on converting to and from moles!

General Steps to Solving a Stoichiometry Problem:

1. Convert the values given in the problem about a reactant or product to a **number of moles**
2. Use the **stoichiometric coefficients** from the **balanced** reaction to **find the number of moles of the unknown** you are being asked for
3. Convert the number of moles of your unknown to a mass, or whatever quantity you are being asked for

Example

2.6g of sodium metal (Na) reacts with water to form NaOH and H₂ according to the unbalanced reaction below. Once the reaction is complete how many grams of NaOH are formed?



WIZE TIP

On an exam, your prof will NOT specify if an equation is balanced or unbalanced.

Always double check that the equation is balanced. If it's not, balance the equation before continuing!

The equation must be balanced in order to get the correct answer!

Step 1: Balance the reaction



Step 2: Find the number of moles of sodium

$$n_{\text{Na}} = \frac{m}{MW} = \frac{2.6 \text{ g}}{22.990 \text{ g/mol}} = 0.113 \text{ mol}$$

Step 3: Find the number of moles of NaOH

$$n_{\text{NaOH}} = n_{\text{Na}} \times \frac{\text{coefficient NaOH}}{\text{coefficient Na}} = 0.113 \text{ mol} \times \frac{2}{2} = 0.113 \text{ mol}$$

Step 4: Convert the number of moles of NaOH to a mass

$$n = \frac{m}{M}$$

$$m_{\text{NaOH}} = n \times MW = 0.113 \text{ mol} \times 39.997 \text{ g/mol} = 4.520 \text{ g} = 4.5 \text{ g}$$

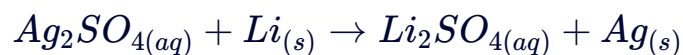
Example: Converting Mass of Reactant to Mass of Product

Calculate the mass of hydrogen gas produced if 0.550 g of iron powder is reacted with an excess amount of sulfuric acid.



Example: Solution Stoichiometry to Determine Mass of Product

Lithium metal was added to a 25mL of a 1.3M solution of Ag_2SO_4 . The unbalanced chemical reaction is shown below.

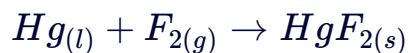


Once the reaction has gone to completion, what mass of silver metal is produced?

- a) 0.0070 g
- b) 7.0 g
- c) 3.5 g
- d) 0.0035 g

Practice: Calculate the Mass of Product

16.0 mL of Hg (mp = -38.8°C , density = 13.56 g/mL) reacts with fluorine gas and the reaction goes to completion as shown below.



Once the reaction has gone to completion, what mass of mercury fluoride, HgF_2 , is produced?

A) 1.48 g

☐

B) 514 g

☐

C) 142 g

☐

D) 258 g

☐

Practice: Stoichiometry Calculations Application

A solution that is 183mM (millimolar) NaCl(aq) is isoosmotic with plasma. This means that cells don't swell or shrink when in this solution. How many grams of sodium chloride are required to make 150mL of isoosmotic NaCl(aq)? The molar mass of NaCl is 58.44g/mol.

☐ 0.2g☐ 1.6g☐ 2.2g☐ 4.9g

7.4 Limiting Reagents

7.4.1

Introduction to Limiting Reagents

Anytime **reactant species are in limited supply** and not present in perfectly proportional amounts, a chemical reaction will have a **limiting reagent** which will be totally **consumed before any other reactant**

The quantity of the limiting reagent available directly **determines the maximum number of product molecules** that can be formed!

Let's Consider an Example:

When making smores (yum!!) the "reaction" looks something like:



If I had 10 graham crackers, 6 pieces of chocolate, and 6 marshmallows, what would be the limiting reagent? In other words, what would I run out of first? And how many smores could I make?

- The "chemistry way" to figure this out would be to **take the # of moles of each reactant and divide by its stoichiometric coefficient**:

Graham crackers: $10/2=5$

Pieces of chocolate: $6/1=6$

Marshmallows: $6/1=6$

- Now, **to figure out the limiting reagent, look at which of the above numbers are the smallest!!**
 - 5 is smallest, therefore Graham crackers are the limiting reagent!!

Now how many smores could we make?

We know graham crackers will determine how much product we get since we'll run out of the crackers first.

$$\text{moles of graham cracker} \times \frac{1 \text{ mol smores}}{2 \text{ moles graham cracker}} = \text{moles of smores}$$

 WIZE CONCEPT

It is necessary to **determine the limiting reagent** whenever we are **given the amounts of two or more reactants** in a chemical reaction.

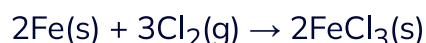
The limiting reagent is **totally consumed in the reaction**. **Any other reactant is in excess**.

Limiting reagent stops the reaction by **running out first**.

The **quantity of the limiting reagent** available directly **determines the maximum number of products** molecules that can be formed

Example: Determine the Mass of Product in a Limiting Reagent Problem

Iron and chlorine gas react to form iron (III) trichloride. If 110 g of iron and 105 g of chlorine gas are reacted, which species is the limiting reagent? What is the maximum mass of FeCl_3 that can be formed?



WIZE TIP

Steps for Solving Limiting Reagent Problems:

Step 1 – Write & **balance** the equation.

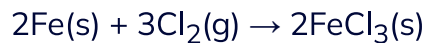
Step 2 – Calculate **moles of each reactant**.

Step 3 – Use the molar ratios of the present reactants (from the balanced equation) to determine the limiting reactant (LR).

- Take the **# of moles of each reactant** from step 2 and **divide by the stoichiometric coefficient** for that reactant
 - **Smallest value** from this calculation tells us the **LR**

Step 4 – From the limiting reactant, use the **molar ratio** (from the balanced equation) to **calculate moles of the desired product**.

Step 5 – Convert moles to the desired units (density, molarity, grams, etc...)



Step 1 – Write & balance the equation.

Is the equation given balanced? _____

Step 2 – Calculate moles of each reactant.

i) Find moles of Fe

ii) Find moles of Cl_2

Step 3 – Use the molar ratios of the present reactants (from the balanced equation) to determine the limiting reactant.

Therefore the limiting reagent is _____ ! (because it had the smallest number)

Step 4 – From the limiting reactant, use the molar ratio (from the balanced equation) to calculate moles of the desired product.

- The maximum mass of FeCl_3 that is formed depends on how much _____ we have (LR).

Step 5 – Convert moles to the desired units (density, molarity, grams, etc...)

Example: What is the Limiting Reagent?

2.0 g of aqueous Barium hydroxide ($\text{Ba}(\text{OH})_2(\text{aq})$, 171.32 g/mol) and 1.5 g of liquid hydrogen bromide ($\text{HBr}(\text{l})$, 80.91g/mol) react together to give aqueous barium bromide ($\text{BaBr}_2(\text{aq})$) and liquid water ($\text{H}_2\text{O}(\text{l})$). Which species would be the limiting reagent of this reaction?

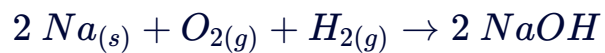
Example: How Many Grams of the Excess Reagent Will Remain?

The reaction between P_4 and Br_2 is very exothermic and produces PBr_5 as the only product. If 7.0 g of P_4 react with 12.0 g of Br_2 how many grams of the excess reagent will remain?

- a) 0.4 g
- b) 4.7 g
- c) 6.1g
- d)11.1 g

Practice: What is the Limiting Reagent

If 12 g of sodium is mixed with 0.64 g of H_2 and 17 g of O_2 which reagent will be limiting in the reaction shown below?



A) Na

☐

B) O_2

☐

C) H_2

☐

D) None, they will all be completely consumed

☐

7.5 Percent Yield

7.5.1

Percent Yield

So far we have been assuming that the reaction proceeds 100% to products. In reality, this is rarely the case!

When we are doing experiments in the lab we often lose some of our product by spilling it or residues of it get left on glassware and spatulas.



So far we have been calculating the **theoretical yield**.

Theoretical Yield - the maximum amount of product produced based on the quantity of the limiting reagent (e.g. the amount of product produced in a calculation).

Actual Yield - the actual amount of product produced in the reaction (e.g. the amount of product obtained in a laboratory experiment).

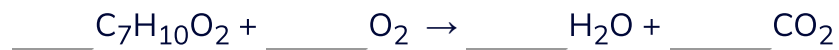
$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

Test Your Understanding

- 1) In a lab, you measure your product and record 5.222g. This is the _____ yield.
- 2) If everything went 100% perfectly according to the reaction we would get the _____ yield
- 3) If I determined the limiting reagent, and then calculated the number of moles of product and the mass of product based on this, I've just calculated the _____ yield

Example: What is the Percent Yield

When 49.00 g of a hydrocarbon fuel with formula $\text{C}_7\text{H}_{10}\text{O}_2$ is reacted with excess oxygen, a total of 21.56 g of water is collected. What was the percent yield of the reaction?



Practice: Calculate Moles Given Percent Yield

If the yield for the following reaction is 45%, how many moles of KClO_3 are needed to produce 1 mol of O_2 ?



A) 0.4 moles

☐

B) 1.1 moles

☐

C) 1.5 moles

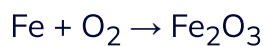
☐

D) 4.6 moles

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 1

What is the limiting reagent in this reaction?

A) Fe

☐

B) O_2

☐

C) Fe_2O_3

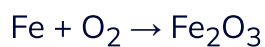
☐

D) There is no limiting reagent

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 2

What is the percent yield?

A) 2.2%

☐

B) 8.8%

☐

C) 9.4%

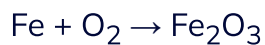
☐

D) 33.1 %

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 3

How many Fe atoms are present in the rust produced?

A) 0.89×10^{17} atoms

☐

B) 1.77×10^{23} atoms

☐

C) 3.56×10^{23} atoms

☐

D) 4.33×10^{23} atoms

☐

7.6 Mass Percentages

7.6.1

Mass Percentage

Shows the amount that each element in a molecule contributes to the overall molecular mass.

WIZE TIP

To determine the mass percentage for a particular element, follow these steps:

- 1) Find the **atomic mass contribution of the element** in question
- 2) Find the **molar mass of the molecule**
- 3) **Solve for the mass percentage** of an element using the equation below (take the mass contribution of an element and divide by the total molar mass, then multiply by 100)

$$\text{mass percentage of element} = \frac{\text{mass controbution of element}}{\text{total mass of molecule}} \times 100$$

Example: What is the percentage of oxygen by mass in sodium hydroxide, NaOH?

Therefore, oxygen contributes to _____ % of the overall mass of NaOH!

Example: Mass Percentages of Elements in A Molecule

Sodium Bicarbonate (NaHCO_3) is a very common chemical frequently used in the food industry. Find the mass percentages of each element in sodium bicarbonate.

Practice: Finding the Mass Percentage of Oxygen in Ethanol

What is the mass percentage of oxygen in ethanol ($\text{C}_2\text{H}_5\text{OH}$)?

A) 31.1 %

☐

B) 34.7 %

☐

C) 35.5 %

☐

D) 41.1 %

☐

7.7 Empirical vs Molecular Formulae

7.7.1

Empirical Vs Molecular Formulas

Molecular Formulas

This is the type of formula we are most familiar with. It tells us **exactly how many atoms make up a molecule**.

Example:

C_6H_6 tells us that for each molecule of C_6H_6 , there are _____ C atoms and _____ H atoms.

Empirical Formulas

If we know the molecular formula, we can find the empirical formula by dividing the number of atoms by the greatest common factor.

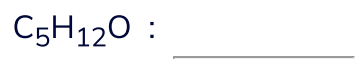
You can think of the empirical formula as the **smallest possible "unit" of the molecular formula**.

Example:

The empirical formula of C_6H_6 would be: _____

Example: Empirical Formulas

What are the empirical formulas of the following molecules?



7.8 Empirical and Molecular Formulae Problems

7.8.1

Example: Empirical and Molecular Formula (Problem Type #1)

(Finding the Empirical and Molecular Formula Given Percent Compositions)

WIZE TIP

To find the empirical and molecular formulae given percent compositions, follow these steps:

Step 1 – Find the **mass %** composition of **each compound**

Step 2 – Assume 100 g sample and **calculate the grams of each element**

Step 3 – Calculate the **moles of each element**

Step 4 – **Divide** each number of moles by the **greatest common factor**

Step 5 – Write out the **empirical formula!**

Step 6 – If given the **molecular weight** of the molecule you can now find the **molecular formula**

- molar mass of empirical formula(x) = molar mass of molecule
- Multiply each subscript in the empirical formula by x to get the molecular formula!

Analysis of an unknown compound tells us that a sample has 39% C, 10% H, and 51% Oxygen. The molar mass of the compound was found to be 124g/mol. What is the empirical and molecular formula for this compound?

Step 1 – Find the mass % composition of each compound

Here we already have been given the percent composition by mass of each element:

C makes up _____%, H makes up _____%, and O makes up _____% of the sample by mass

Step 2 – Assume 100 g sample and calculate the grams of each element

If we assume a 100g sample, how many grams would be accounted for by C, H, and O?

C= _____ g

H= _____ g

O= _____ g

Step 3 – Calculate the moles of each element $n = \frac{m}{M}$

Step 4 – Divide each number of moles by the greatest common factor

Step 5 – Write out the empirical formula!

Now we know that for every _____ C, there are _____ H and _____ O

Based on this information, we can write the empirical formula: _____

Step 6 – If given the molecular weight of the molecule you can now find the molecular formula

- We are told the molar mass of the whole molecule is: 124g/mol
- If we find the molar mass of the empirical formula, we can see if we need to multiply it by a certain number to get the molar mass of the overall compound:

Example: Empirical And Molecular Formula (Problem Type #2)

(Finding the Empirical and Molecular Formulae Given Masses in Grams of Elements)

A sample of a compound contains 1.52g of N atoms and 3.47g of O atoms. The molar mass of the compound is between 90.0g and 95.0g. Determine the empirical and molecular formulas. Also, calculate the actual molar mass of this compound.

7.8.3

Practice: Molecular Formula from Empirical Formula

p-dichlorobenzene ($M = 147.00 \text{ g/mol}$) is a common disinfectant and deodorant used in industrial applications. It has an empirical formula of $\text{C}_3\text{H}_2\text{Cl}$. Determine its molecular formula.

Note: if your answer is $\text{C}_3\text{H}_2\text{Cl}$, enter it as: C3H2Cl.

Answer

Practice: Determine the Empirical and Molecular Formulae

A compound is found to contain 50.05% sulfur and 49.95% oxygen by weight. The molecular weight for this compound is 64.07 g/mol.

Part 1

What is the empirical formula for this compound?

A) SO

☐

B) S₂O

☐

C) SO₂

☐

D) SO₃

☐

Practice: Determine the Empirical and Molecular Formulae

A compound is found to contain 50.05% sulfur and 49.95% oxygen by weight. The molecular weight for this compound is 64.07 g/mol.

Part 2

What is the molecular formula for this compound?

A) S₂O₂

☐

B) SO₄

☐

C) S₂O

☐

D) SO₂

☐

7.9 Combustion Reactions + Empirical and Molecular Formulae

7.9.1

Introduction to Combustion Reactions

Combustion reactions: A reaction between a hydrocarbon ($C_xH_yO_z$) with oxygen to produce carbon dioxide (CO_2) and water (H_2O).



Combustion reactions are commonly used in elemental analysis to determine the empirical formula of a compound.

Steps to Balance a combustion reaction:

1. Balance the carbons
2. Balance the hydrogens
3. Balance the oxygens using ONLY the coefficient of elemental oxygen (O_2)
4. If the equation has any fractions (often times the coefficient in front of O_2 is a fraction), multiply the ENTIRE equation by 2

Example: Balancing the Following Reaction



Determining Empirical Formulae from Combustion Reactions

Steps to Calculate Empirical Formulae:

1. Determine the **moles of carbon dioxide**, $n(\text{CO}_2)$. This will **give you the moles of carbon** since $1\ n(\text{CO}_2) = 1\ n(\text{C})$.
2. Determine the **moles of water**, $n(\text{H}_2\text{O})$. This will give you **moles of hydrogen** since $1\ n(\text{H}_2\text{O}) = 2\ n(\text{H})$.
3. Calculate the **mass of carbon and the mass of hydrogen** using their molar masses.
4. Calculate the **mass of oxygen** by subtracting the masses of carbon and hydrogen from the mass of the hydrocarbon
5. Calculate the **moles of oxygen** using the mass of oxygen and its molar mass.
6. With the moles of C, H, and O known, **divide all 3 mole values by the LOWEST value**. If there are any fractions, multiply by a common factor to get whole numbers.

Example: Determine the Empirical Formula from a Combustion Reaction

A 55.4 g sample of a compound containing only C, H and O was burned in the air to produce 143.2 g of carbon dioxide and 68.4 g of water. Determine the empirical formula of this compound.

Practice: Combustion Analysis Calculation

When 2.11 g of an unknown hydrocarbon (compound made up of only carbon and hydrogen atoms) was burned in excess oxygen as part of a combustion analysis. In the process 6.32 g of CO_2 and 3.45 g of H_2O are produced. What is the empirical formula of the hydrocarbon?

A) C_2H_6

☐

B) C_3H_8

☐

C) CH_4

☐

D) C_6H_6

☐